

What Is Artificial Intelligence?

Lecture 1

CS 4880

Kirk Duran

Artificial Intelligence

Key idea

AI has more public attention than ever. Before we study the techniques, we need to decide what we actually mean by “artificial intelligence.”

- ▶ AI is no longer confined to science fiction or research laboratories.
- ▶ It increasingly shapes work, education, medicine, media, science, and everyday decisions.
- ▶ The term “AI” now refers to many different systems with very different capabilities.

In class

When you hear “AI,” what system or capability comes to mind first?

How We Will Talk In This Course

Socratic Method

We will often begin with a question before we introduce the formal answer.

- ▶ I will ask for your intuition.
- ▶ I may call on people around the room.
- ▶ “I do not know” is a valid starting point.

Key idea

The goal is not to guess what I am thinking. The goal is to expose assumptions, compare explanations, and make our reasoning precise.

Today

- ▶ What do we mean by artificial intelligence?
- ▶ What do we mean by intelligence?
- ▶ Where did AI come from?
- ▶ What kinds of methods count as AI?
- ▶ What questions will this course try to answer?
- ▶ What larger technical and social questions surround the field?

Part 1: What Is Artificial Intelligence?

Two horizontal lines are positioned below the title. The top line is orange and shorter, while the bottom line is white and longer, extending across most of the slide width.

A Working Definition

What should count as AI?

In class

Propose a one-sentence definition. What must a system be able to do before you would call it intelligent?

Definitions Emphasize Different Things

AIMA

Focuses on **agents**: systems that perceive an environment and choose actions.

Industry

Often emphasizes machines performing tasks associated with human reasoning, prediction, or decision-making.

Engineering

Often frames AI as building systems that exhibit useful intelligent behavior.

Key idea

There is no single universally accepted definition. Different definitions emphasize behavior, mechanisms, goals, or comparison with humans.

More Ways To Define AI

Human-like Tasks

Systems that perform tasks normally associated with human intelligence: perception, language, reasoning, and decision-making.

Goal-Directed Systems

Machines that make predictions, recommendations, or decisions in pursuit of human-defined objectives.

Learning And Problem Solving

Computational systems that learn, reason, search, plan, or otherwise solve problems.

Our Working Definition

Definition

Artificial intelligence is the study and engineering of computational systems that perceive, reason, learn, decide, or act in ways that accomplish goals we associate with intelligent behavior.

- ▶ The system does not need to think exactly like a human.
- ▶ The system does not need to be generally intelligent.
- ▶ Intelligence can appear in narrow, well-defined tasks.
- ▶ We care about both **what** the system can do and **how** it does it.

Why Are You Studying AI?

I want to hear from you.

In class

What made you enroll in this course? Curiosity, career goals, research, fear of missing out, wanting to understand modern systems, something else?

Why I Am Interested In AI

Automation

If a machine can reliably do repetitive work, why should a human have to do it?

Optimism

AI can amplify human capability and potentially improve health, science, accessibility, education, and productivity.

Philosophy

Building artificial minds forces us to ask what intelligence is, what understanding is, and what is uniquely human.

Part 2: Where Did AI Come From?



A Very Short History

Key idea

AI has repeatedly moved through cycles of ambitious goals, technical breakthroughs, bottlenecks, disappointment, and renewed progress.

- ▶ **1950s–1960s:** symbolic reasoning, search, theorem proving.
- ▶ **1970s–1980s:** expert systems and knowledge engineering.
- ▶ **1990s–2000s:** probabilistic reasoning, data-driven learning, stronger computing infrastructure.
- ▶ **2010s–present:** deep learning, large-scale models, reinforcement learning, and generative AI.

The Recurring Question

What parts of intelligent behavior can be represented computationally?

Key idea

The tools change, but AI keeps returning to this question.

20th-Century Beginnings

Bold Ambition

Build machines that could reason and act in ways associated with human intelligence.

Scientific Motivation

Formalize reasoning, problem solving, language, learning, and decision-making.

Engineering Motivation

Automate difficult tasks and extend what machines can do.

Social Context

War, economic competition, research funding, and technological optimism influenced the field's priorities.

1950: Turing Changes The Question

- ▶ Alan Turing asked whether machines could exhibit intelligent behavior.
- ▶ He proposed an operational test based on text conversation.
- ▶ The focus shifts from defining an invisible mental property to examining observable behavior.

Question

If a machine can convincingly imitate intelligent conversation, is that enough to call it intelligent?

Key idea

When a concept is hard to define directly, ask what evidence would demonstrate it.

1956: Artificial Intelligence Becomes A Field

- ▶ John McCarthy helped organize the Dartmouth summer research project where the term **artificial intelligence** was introduced.
- ▶ Researchers proposed that aspects of learning and intelligence might be described precisely enough for machines to simulate them.
- ▶ The field began with unusually ambitious expectations about general-purpose machine intelligence.

In class

Is it easier to build a machine that is excellent at one task, or a machine that is broadly competent across many tasks? Why?

Part 3: What Do We Mean By Intelligence?

Two horizontal lines are positioned below the title. The top line is orange and shorter, while the bottom line is white and longer, extending across most of the slide width.

What does intelligence mean?

In class

Name one capability you think is essential to intelligence. Could a system possess that capability without possessing the others?

Behavior Is Not The Same As Understanding

The Chinese Room

John Searle's thought experiment asks us to imagine a person manipulating unfamiliar symbols by following formal rules.

- ▶ Inputs arrive as symbols.
- ▶ Rules determine which symbols to return.
- ▶ To an outsider, the responses may appear meaningful.

Key idea

The philosophical challenge: can correct symbol manipulation produce genuine understanding, or only the appearance of understanding?

In class

Does this argument become more or less convincing when applied to modern language models?

AI Is More Than ChatGPT

Key idea

Large language models are one part of AI. The field includes systems with very different goals, representations, and mechanisms.

Simple Automation

Sensors, rules, controllers, and reactive systems.

Modern AI

Search, planning, probabilistic models, machine learning, neural networks, robotics, and language systems.

General Intelligence

Hypothetical systems with broad, flexible competence across many domains.

AI Is A Toolbox Of Mechanisms

Search	Logic	Probability	Learning	Hybrid Systems
Explore possible states or actions.	Represent facts and derive conclusions.	Reason under uncertainty.	Improve behavior from data or experience.	Combine multiple representations and methods.

Key idea

AI is not one algorithm. It is a collection of approaches for building systems that behave intelligently.

Part 4: What Will We Study?

Two horizontal lines are positioned below the title. The top line is orange and shorter, while the bottom line is white and longer, extending across most of the slide width.

AI Course Roadmap

Unit 1: Intelligent Agents

Perception, actions, environments, rational behavior.

Unit 2: Search

State spaces, heuristics, path finding, optimization.

Unit 3: Adversarial Games

Minimax, alpha-beta pruning, Monte Carlo

Unit 4: Knowledge-Based Systems

Logic, representation, inference, planning.

Unit 5: Uncertain Knowledge

Probability, Bayes, graphical models, tracking.

Unit 6: Learning

Machine learning, neural networks, reinforcement learning.

Unit 7: Language-Based Systems

Language models and modern generative AI.

Questions We Hope To Answer

- ▶ How can an agent perceive its surroundings and choose useful actions?
- ▶ How can a machine explore alternatives through trial and error?
- ▶ How can a program search large spaces efficiently?
- ▶ How can computers represent knowledge and draw conclusions?
- ▶ How can AI reason when information is uncertain?
- ▶ What allows a system to learn from data or experience?
- ▶ How do modern language and generative systems work at a high level?
- ▶ Which problems remain fundamentally difficult?

The Ambitious Goal

How general can artificial intelligence become?

Artificial General Intelligence

Definition

Artificial General Intelligence (AGI) usually refers to a hypothetical system with broad, flexible competence across many intellectual tasks rather than exceptional performance on only one narrow problem.

- ▶ Modern AI is impressive but uneven: systems can be extraordinary in one setting and brittle in another.
- ▶ AGI is a research aspiration and a contested concept, not a capability we should assume has already been achieved.
- ▶ Studying narrow AI still teaches us the mechanisms from which more general systems may be built.

Part 5: What Should We Want From AI?



Is This A Good Thing Or A Bad Thing?

I want your opinion.

In class

If AI becomes dramatically more capable over your lifetime, what is one outcome you hope for and one outcome you worry about?

Automation, Power, And Control

The Promise

Automation can reduce repetitive labor, improve access to expertise, accelerate discovery, and expand human capability.

The Risk

The same systems can concentrate decision-making power, enable surveillance, encode bias, or make institutions harder to challenge.

Example

A useful political-philosophy lens: systems can trade autonomy for coordination, efficiency, or security.

In class

Who should be allowed to delegate decisions to AI, and which decisions should remain meaningfully contestable by humans?

Case Study Prompt: Replacing Human Work

Key idea

Automating a task is not the same as automating a job, and replacing a worker is not the same as replacing the judgment surrounding the work.

- ▶ Which parts of a job are predictable enough to automate?
- ▶ Which parts depend on context, accountability, trust, or tacit knowledge?
- ▶ What happens when organizations automate faster than they understand the failure modes?
- ▶ If automation fails, who absorbs the cost?

Part 6: Bigger Questions

Two horizontal lines are positioned below the title. The top line is orange and shorter, while the bottom line is white and longer, extending across most of the slide width.

The Search For The Simplest Life Form

Biology's Question

What is the smallest set of mechanisms required for something to remain alive, reproduce, and adapt?

- ▶ Minimal genomes.
- ▶ Synthetic cells.
- ▶ Core mechanisms versus accumulated complexity.

AI Analogy

What is the smallest collection of mechanisms required for behavior we would call intelligent?

In class

Is there a “minimum viable mind”?
What capabilities would have to be included?

Mapping The Brain

Neuroscience

Researchers map biological brains at increasingly fine levels: regions, cell types, neurons, and connections.

- ▶ The brain is extraordinarily complex.
- ▶ Even small nervous systems can be difficult to fully characterize.

Question For AI

Does understanding biological intelligence tell us how to engineer artificial intelligence?

- ▶ Copy the mechanism?
- ▶ Reproduce the function another way?
- ▶ Learn principles without copying biology literally?

Limits Of Formal Systems

Gödel's Incompleteness Theorems

Roughly: sufficiently expressive formal systems contain statements whose truth cannot be established using only the system's own proof rules, assuming the system satisfies the required conditions.

Key idea

This is a theorem about formal mathematical systems, not a theorem that proves human minds are non-computational or that AI has a particular ceiling.

Limits Of Formal Systems: Discussion

In class

Why are arguments about the limits of formal reasoning so tempting in discussions of intelligence? What additional assumptions would be needed to connect them to minds?

Part 7: Course Logistics



Course Logistics And Syllabus

Canvas

Announcements, assignments, grades, due dates, and course documents.

Lectures

Slides are a scaffold. Class discussion, worked examples, and questions are part of the course content.

In class

Open the syllabus on Canvas. We will walk through grading, late work, collaboration, AI-tool policy, labs, and major deadlines.

About Me

- ▶ Full-Time Lecturer in Computer Science and Software Engineering.
- ▶ Education in applied mathematics, computer science, and computer engineering.
- ▶ Research interests: deep reinforcement learning, heuristic search, robotics, and computer science education.
- ▶ Industry experience in robotics engineering and machine learning engineering.
- ▶ Mentoring through Beacons, OUDI, CENG SURP, research projects, and career preparation.

Outside Class

Basketball, weightlifting, video games, anime, and thinking too much about computers.

Key idea

I am here to help you learn the material and navigate the field.

Community

Color Coded by ColorStack

A Cal Poly community centered on computing, belonging, peer support, professional development, and access to opportunities.

- ▶ Join the Discord.
- ▶ Follow @colorcodedslo.
- ▶ Watch for events, resources, and opportunities.
- ▶ Bring a friend.

Waitlist And Roll Call

- ▶ If you are on the waitlist, keep attending while enrollment settles.
- ▶ I will communicate what I can do as seats and department constraints become clear.
- ▶ Make sure I know you are here and that your enrollment status is correct.

Before Next Time

- ▶ Read the syllabus.
- ▶ Confirm you can access Canvas and the course materials.
- ▶ Bring one example of a system you think behaves intelligently.
- ▶ Be ready to defend why you think it qualifies as AI.

Key idea

Next time: intelligent agents, environments, percepts, actions, and rational behavior.

Next Time

Intelligent Agents

In class

Join me in lab.